

ARIMA model selection for UK inflation time series forecasting

Introduction

Inflation affects the purchasing power of households, business costs, and wider macroeconomic stability, therefore accurate inflation forecasting is very important for all economic agents and especially policymakers. Households and consumers make decisions to hedge themselves for the future under these conditions. These decisions have consequences on economic growth, unemployment, consumer and business confidence, wages etc and therefore it is in the interest of the central bank to forecast inflation in order to inform policy decisions that prevent fluctuations in economic activity which negatively impacts social welfare.

Mihalache et al., 2023 highlights the role of monetary authorities in tackling these price fluctuations via changes in the Bank Rate and quantitative easing/tightening (Mihalache et al., 2023). The Monetary Policy Committee (MPC) under the Bank of England (BoE) aims to maintain a 2% target level of inflation. Forecasting inflation means the MPC can implement successful policy to maintain this baseline and prevent the adverse effects discussed. One method is to model inflation as a time series by following the repeating dynamics over time. Econometric models can map this pattern to forecast inflation for the future, aiding the policy decisions of central banks.

However, problems with future forecast arise as inflation is extremely volatile. As inflationary pressure is stimulated via both the demand and supply side of the economy, it is easily susceptible to unpredictable and long-lasting shocks e.g. the Ukraine Russia war, Covid-19, and the 2008 financial crisis. The BoE emphasize that forecasting frameworks are known to fail especially in the aftermath of external inflationary shocks and therefore banks prioritise a combination of models in policy making. (Baldwin et al., 2023)

A common econometric model in time series forecasting is the ARIMA model. This report aims to fit the best performing ARIMA (p,d,q) model to UK monthly inflation data over the past 30 years. Performance is reviewed across the following metrics: AIC, BIC, RMSE, and MAE. The forecast over the testing period will be compared with real life to evaluate the accuracy of the model and its applicability in real life.

Literature review:

The importance of inflation forecasting:

The BoE uses macroeconomic forecasts to guide monetary policy decisions, building a framework for the outlook on inflation whilst holding individual MPC members accountable for policy decisions. This is important as the BoE can guide inflation expectations in financial markets for households and firms to influence behaviour in support of economic objectives (Baldwin et al., 2023).

However, over time inflation has become increasingly harder to forecast which poses a threat to policymakers. Stock et al. (2007) uses US GDP price index quarterly data

to explain this. The inflation time series is built up of a permanent trend and a serially uncorrelated transitory component. The variance of the permanent disturbance has experienced large changes, but the variance of the transitory disturbance has remained essentially the same. Therefore, with the dynamics changing over time due to structural economy changes, forecasts deteriorate (Stock and Watson, 2007). On the other hand, current inflation has become less volatile over time, making inflation easier to forecast as the risk measured by mean squared forecast errors has decreased (Stock and Watson, 2007).

Phillips curves for forecasting:

Another approach for inflation forecasting is the Phillips curve. Atkeson et al., 2001 uses the non-accelerating inflation rate of unemployment (NAIRU) Phillips Curve model to compare accuracy across 3 sets of inflation from 1985-2000 with forecasts that suggest inflation will remain constant – aka the naïve forecast. The study concludes the Phillips Curve approach is no more accurate than the naïve forecast across all 3 data sets (Atkeson et al., 2001).

This however has led to questions on the feasibility of inflation forecasts based on the unemployment rate as the relationship between these variables changes with the economic environment (Atkeson et al., 2001).

The ARIMA model:

The ARIMA model is an alternative forecasting tool based on econometric models, proving to successfully forecast even highly volatile time series. Hyndman and Khandakar (2008) used the ARIMA model with a package in R studio for future forecasting. They identified the order of differencing (d) using a unit root test and the order for the AR (p) and MA (q) series by using the AIC measure. This is different to this report where ACF and PACF plots are used to identify p and q. They highlight that choosing the order of differencing by minimizing the AIC can lead to over differencing which is harmful for forecasting as this can widen prediction intervals. However, although unit root tests are favoured, most are based on the hypothesis that unit roots exists which can bias results towards more differences (Hyndman and Khandakar, 2008).

Hyndman and Athanasopoulos (2021) discuss the construction process for the ARIMA model. Firstly, the time series must hold necessary statistical properties to be stationary. They describe the differencing process that transforms a non-stationary time series into a stationary one by eliminating trends and seasonality, further discussing the KPSS unit root test to identify a stationary time series. Alternatively, they discuss the ACF function as an unofficial method of deriving stationarity. Under the general condition, the ACF function should quickly drop to 0 if the time series is stationary in contrast to a non-stationary time series that decays slowly (Hyndman and Athanasopoulos, 2021).

Curiously, this contradicts the data found in this report. Although the ADF test provides evidence against a unit root, the ACF displays relatively slow decay which illustrates that graphical tests and formal statistical tests may not always provide identical signals and instead the ADF test was favoured when determining the order of differencing. Furthermore, Hyndman and Athanasopoulos (2021) continue to discuss the AR and MA models that combine with differencing to create the non-seasonal ARIMA model, and how the ACF and PACF can be used to determine appropriate values for p and q as conducted in this study (Hyndman and Athanasopoulos, 2021).

Real life inflation forecasts:

The ARIMA model has been applied to some real-life economies. An example is Mihalache's 2023 paper on forecasting inflation in Romania where the ARIMA (7,1,1) model was found suitable for the sample period under model identification, parameter estimation and diagnostic checking (Mihalache et al., 2023).

The process for selection of the ARIMA model in the paper is consistent with the contents of this report starting with the use of the conventional Box-Jenkins method. A unit root test is conducted for stationarity under the ADF, and the order of the AR and MA is identified using ACF and PACF plots. Performance was analysed based on AIC and BIC measures and diagnostic tests were used to analyse the statistical properties via specification error, residual normality, serial correlation and heteroskedasticity. Finally, the MAE and RMSE confirmed model suitability (Mihalache et al., 2023).

Although a suitable model is found, the paper suggests that machine/deep learning methods can improve future forecasting accuracy, noting this comes at the risk of efficiency vs interpretability (Mihalache et al., 2023).

Limitations in future forecasting:

Whilst the ARIMA model has proven a useful forecasting tool, it is important to consider the practical limitations. Petrica et al. (2016) investigates the limitations of the ARIMA model in a financial context, discovering 2 important features in the analysis of financial time series. Although Petrică et al. focus on financial time series, their discussion of extreme events and changing volatility is relevant when considering the challenges of forecasting a macroeconomic variable subject to large shocks. Firstly, 'fat tails' means that extreme or unexpected events occur with a much higher probability under the normal distribution. Secondly, 'volatility clustering' suggests the ARIMA fails to model shocks under the linear variance assumption for stationarity (Petrică et al., 2016).

Furthermore, Hyndman and Athanasopoulos (2021) discuss the implications of the order of differencing for prediction intervals in forecasting. They highlight that for a stationary time series with $d=0$, prediction intervals will all be the same whereas for a non-stationary time series prediction intervals will continue to widen as the forecast

horizon increases. This is problematic for forecasting as statisticians may have low faith in predictions and it could be unwise to use for policy decisions. Whilst this doesn't apply to this study since the time series is found to be stationary, this is uncommon for most financial data (Hyndman and Athanasopoulos, 2021).

The BoE remarks that using all historical time series data is unwise with the 'Lucas critique' highlighting that structural relationships are vulnerable to instability through regime changes. Therefore, standard practise restricts estimation samples to more recent periods. However, there is also concern if sufficient periods are included with large shocks such as Covid-19 occurring that skews predictions (Baldwin et al., 2023).

Furthermore, the BoE emphasizes that although economic models for forecasting are useful tools, policy decisions also rely on a large degree of judgement from committee members. They caution to relying on a single model when it comes to capturing economic shocks. Therefore, banks generally use a combination of models and cross check their implications before implementing policy (Baldwin et al., 2023).

Data

UK monthly inflation data from 1982-2022 is collected from a Kaggle dataset: <https://www.kaggle.com/datasets/scarfsman/uk-inflation-data-1989-2022>. This consists of 415 observations with no missing values and therefore no amendments have been made to the data. Inflation is measured on a CPIH basis, defined as the percentage change in the price level including costs relating to housing which are mainly rent and mortgage payments without considering the cost of housing.

Methodology

ARIMA model:

The ARIMA model (Auto Regressive Integrated Moving Average model) is a time series analysis model for future forecasting. Past time series data is analysed to find trends, cycles, and seasonality in order to predict future values. The ARIMA model integrates the Auto Regressive (AR) and Moving Average (MA) time series models.

The data is divided into a training and test set. The model is estimated using the training data, meanwhile the test set is reserved for evaluating out-of-sample forecasting performance. In this report, the ARIMA (p,d,q) will be trained on current inflation data and used on the test set to assess predictive accuracy. This will be done by splitting the data into a 80:20 train/test split proportion.

Multiple ARIMA (p,d,q) models will be tested and evaluated across 4 performance metrics including: AIC, BIC, RMSE, and MAE.

AR and MA components:

An AR(p) is an autoregressive model of order p. It contains lagged y_t terms (up until lag p) as predictors and ε_t which is a white noise term.

$$AR(p): y_t = \beta_0 + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \dots + \beta_p y_{t-p} + \varepsilon_t$$

An MA(q) is a moving average model of order q. It contains lagged error terms up until lag q and the original error/ white noise term ε_t .

$$MA(q): y_t = \alpha_0 + \alpha_1 \varepsilon_t + \alpha_2 \varepsilon_{t-1} + \alpha_3 \varepsilon_{t-2} + \dots + \alpha_{q+1} \varepsilon_{t-q}$$

Stationarity:

The time series must be stationary in order to fit the ARIMA model. Stationarity implies that the mean and variance of the data are constant and finite over time as well as the autocorrelation between 2 data points remaining constant. Non-stationary time series must be differenced until they become stationary.

Differencing removes seasonal structures and trends which are common within economic data and past financial crises may have persistence in predictions by autoregressive models and therefore in future predictions made by the ARIMA model.

An Augmented Dickey Fuller (ADF) can be used in order to test for the presence of a unit root in the time series. Under presence of a unit root, the time series isn't stationary and therefore requires differencing.

ACF and PACF:

After stationarity is satisfied, an ARIMA(p,d,q) can be fit where p is the order of AR, q is the order of MA and d is the number of differences required for stationarity. The order of the AR and MA models can be determined using ACF (Autocorrelation function) and PACF (Partial autocorrelation function) plots. The number of MA terms can be assessed by looking at the ACF and the number of AR terms by looking at the PACF.

The autocorrelation is how the time series is correlated with its past values. The ACF measures how the current value is correlated with past values. The x axis represents the number of lags, and the y axis shows correlation strength. Lags show how far back in time we look e.g. lag 1 is the value for inflation 1 month ago, lag 2 is 2 months ago and so forth. Spikes outside the confidence band indicate there is high correlation. The common interpretation is that if correlations cut off quickly, MA behaviour is present. Whereas, if correlations decay slowly, the series suggests there is a possibility for non-stationarity although this isn't conclusive evidence, and a unit root test is favourable as evidence.

Hyndman and Athanasopoulos (2021) explain that the ACF shows autocorrelations between y_t and y_{t-k} . However, if y_t and y_{t-1} are correlated, y_t and y_{t-2} may only be

correlated through y_{t-1} rather than any additional information that y_{t-2} provides. Therefore, partial autocorrelations measure the relationship between y_t and y_{t-k} whilst removing the effects of lags 1 to $k-1$ (Hyndman and Athanasopoulos, 2021).

The PACF removes the influence of intermediate lags and instead answers whether a given lag matters directly towards the current value. For instance, a lag that cuts off at k suggests that only the past k values matter directly. The common interpretation here is that if PACF cuts off it indicates an AR component.

A key point is that lag 0 should typically be ignored because this shows the correlation of the current value with itself (variance), which will always be 1 and therefore provides no additional information.

Model evaluation:

The best ARIMA model amongst the candidates will be chosen by comparing the following measures.

The AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion) evaluate model accuracy whilst accounting for the trade-off with model complexity – where a model with too many parameters may overfit and therefore these models penalize on this basis.

The MAE (Mean Absolute Error) and RMSE (Root Mean Square Error) on the other hand are used for forecast evaluation. Results from each model are evaluated based on prediction error – how close the model predictions are to actual values over the test set.

The best fitting model will have the lowest values across all metrics. Finally, to confirm the chosen model is in fact the best fit, residual errors and an ACF of errors is visualized to check for overfitting. The best fitting model will satisfy the necessary conditions for the residuals.

Results:

Testing for stationarity:

Firstly, a Dickey Fuller test is used to test for stationarity. The null hypothesis of a unit root is tested against the alternative of no unit root.

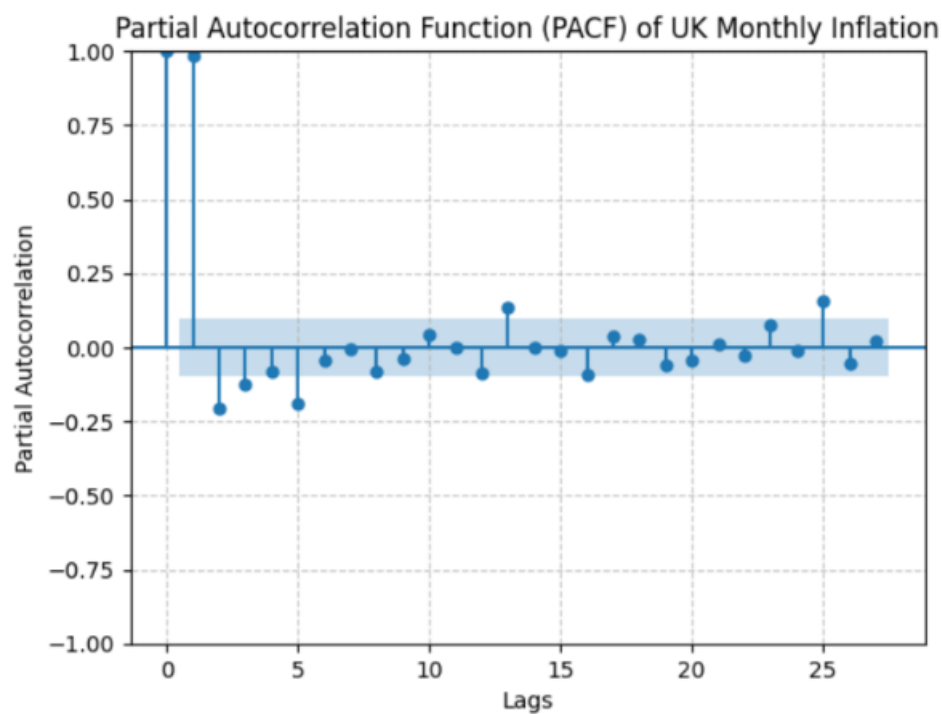
ADF:

```
Results of Dickey-Fuller Test:
Test Statistic      -3.987353
p-value             0.001478
#Lags Used          18.000000
Number of Observations Used  396.000000
Critical Value (1%)  -3.446972
Critical Value (5%)  -2.868866
Critical Value (10%) -2.570673
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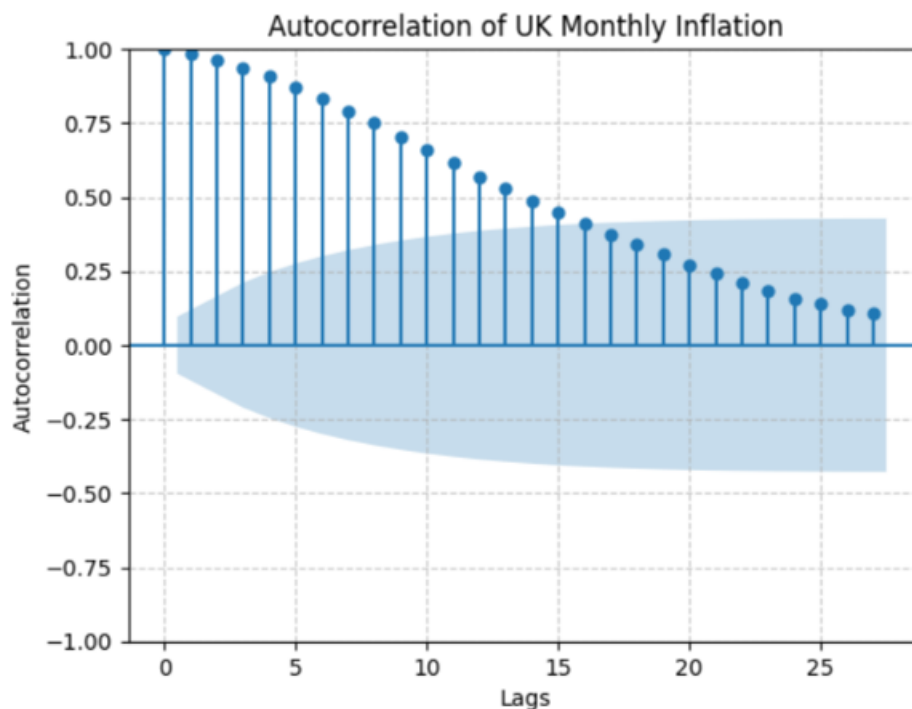
From the results above, the p-value of $0.001478 < 0.05$ and therefore the null hypothesis is rejected. Additionally, the test statistic is more negative than the critical values at the 10%, 5%, and the 1% level. Therefore, it can be determined with confidence that the time series is weakly stationary and no differencing is required. For the ARIMA model, this suggests that $d=0$ is likely appropriate.

Determining the order of AR and MA components:

PACF:



ACF:



From the PACF graph, it is evident the time series cuts off after 2 lags, therefore suggesting there are two AR terms in the model with $p=2$.

However, from the ACF graph, there appears to be gradual decay in the model and therefore this doesn't provide a clear indication of the appropriate MA order. Whilst a slow decaying ACF can be indicative of non-stationarity, the Dickey Fuller test has proven at a low significance level that the time series is in fact stationary and therefore the ADF result is prioritised.

An initial prediction based on conjecture is that ARIMA (2,0,0) where $q=0$ (no MA terms) could be appropriate based on the visualizations. However, testing a variety of model candidates is worthwhile here as the order of MA is inconclusive. Additionally, ACF and PACF plots aren't necessarily concluding evidence for the best model specification. Therefore, several candidate specifications have been estimated and compared empirically.

A comparison will be conducted on 5 ARIMA (p,d,q) models including:

- ARIMA (2,0,0)
- ARIMA (1,0,0)
- ARIMA (1,0,1)
- ARIMA (2,0,1)
- ARIMA (2,0,2)

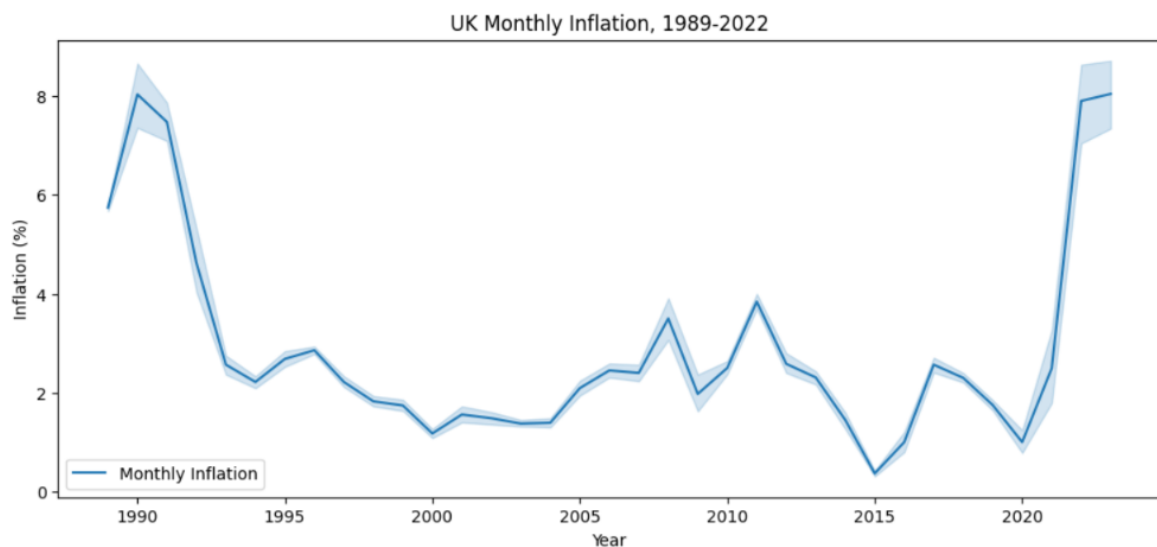
Exploratory Data Analysis:

Summary statistics:

Inflation	
count	415.00
mean	2.84
std	2.05
min	0.20
25%	1.60
50%	2.30
75%	3.00
max	9.60

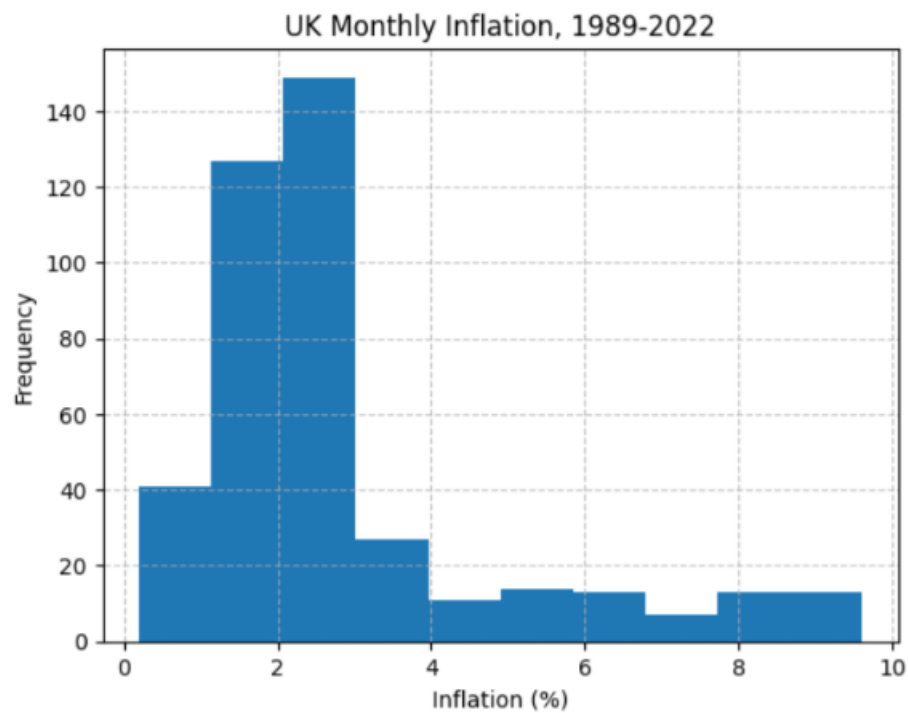
The average value for inflation over the 33 years was 2.84%. this is close to the 2% target average for the UK. However the maximum reached 9.6% and the minimum was 0.2%.

Inflation plot:



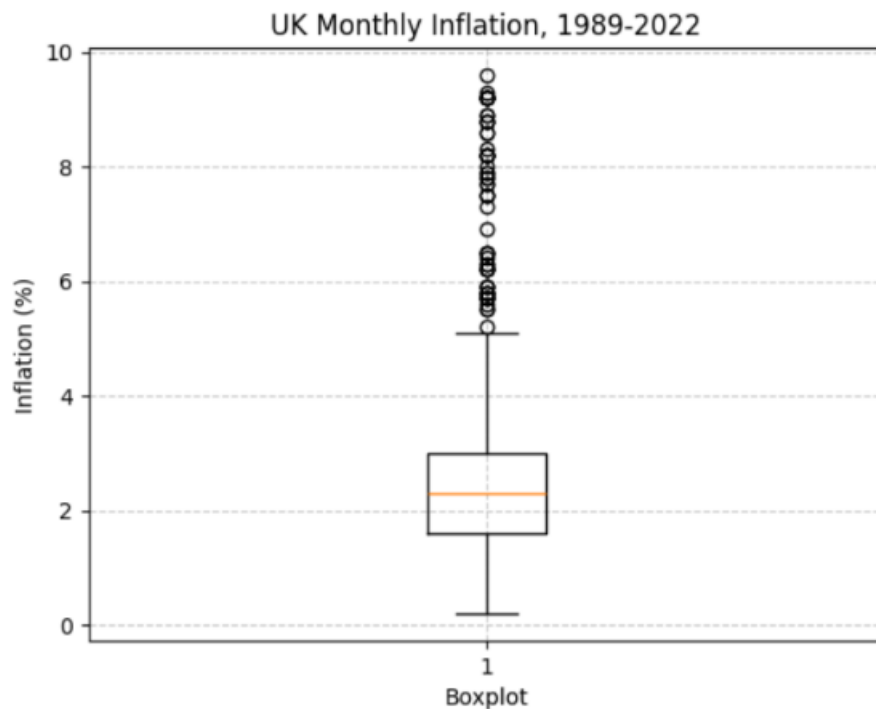
The line graph shows that between 1995-2020, inflation remained relatively stable with a larger spike surrounding the 2008 financial crisis. The larger outliers are observed in 1990 and 2021. The 2021 period showcases the Corona virus pandemic where inflation reached its peak at 9.6%.

Histogram:



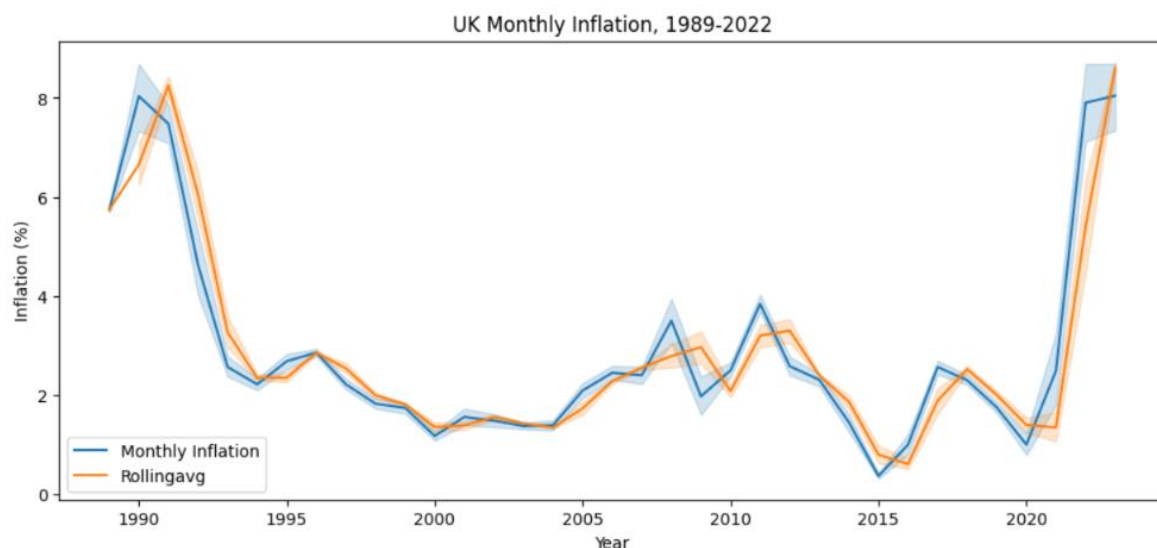
The histogram shows the distribution in frequency of different inflation rates. The most frequent over 150 months was an inflation rate between 2-3%, aligning with the BoE target rate. However, there are many months where inflation was far from this target although less frequent.

Boxplot:



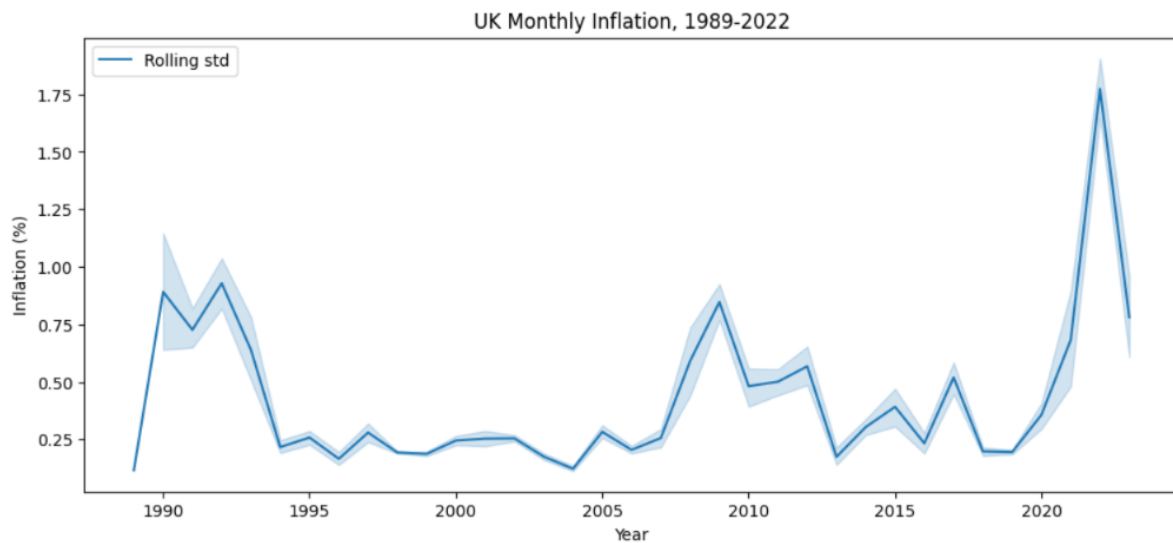
The box plot suggests the average inflation was just above 2% with a minimum just above 0. This differs from the summary statistics as it suggests the peak was around 5% with any values above being classified as outliers from unprecedented shocks.

Rolling mean:



The rolling average is created over a 12-month period in order to eliminate monthly fluctuations and highlights longer-term movements in inflation. There is a sharp discrepancy between the rolling mean and observed inflation during periods of rapid change for example from 2020-2023 which indicates increased volatility.

Rolling variance:



The rolling standard deviation measures how data fluctuates over time (again over a 12-month window). Volatility remains for the periods remarked before, however there was greater volatility surrounding periods of rapid change especially under periods of extreme economic shocks.

Model comparison table:

	Model	AIC	BIC	RMSE	MAE
0	ARIMA(2,0,0)	18.212	33.433	2.981	1.971
1	ARIMA(1,0,0)	29.213	40.629	3.123	2.076
2	ARIMA(1,0,1)	20.058	35.278	3.023	2.003
3	ARIMA(2,0,1)	13.257	32.283	2.764	1.863
4	ARIMA(2,0,2)	22.184	45.015	2.993	1.981

Under the following evaluation metrics, the ARIMA (2,0,1) model performed the best amongst the tested candidates as it scored the lowest values across all measures. The predicted ARIMA (2,0,0) model came in second, suggesting there was indeed a MA term that provided a better fit to the model which wasn't evident from the ACF graph.

AIC/BIC:

The 2 most reliable metrics in ARIMA model selection are AIC and BIC. The AIC measure evaluates and compares across models to select the optimal model without overfitting. However, the AIC provides a trade-off between data accuracy and the number of parameters used as a model with too many parameters can become too complex and overfit. The AIC is therefore the preferred because it estimates the model fit whilst penalizing complexity.

Although both models highlight this trade off, the BIC measure however penalizes complexity to a higher extent than AIC for larger sample sizes; BIC prefers simpler models.

The optimal model will have comparatively lower AIC and BIC values. Determining when to use AIC or BIC depends on the goal of analysis. AIC tends to favour predictive performance whereas BIC imposes a stronger complexity penalty. Therefore, AIC could be preferable when looking to forecast performance and BIC for determining actual data structure or constructing a more interpretable model although this isn't a rule. In this case ARIMA (2,0,1) performed best amongst the candidates for both measures.

Hyndman and Khandakar (2008) however outline certain problems with the AIC and BIC information criteria. They highlight that whilst AIC is useful for selecting predictors and determining the model order, these struggle with selecting the order of differencing as differencing changes the data on which the AIC is computed. Consequently, models with different orders of differencing aren't comparable which affects model evaluation (Hyndman and Khandakar, 2008). However, whilst this is a general limitation, all 5 models in this report have the same order of differencing and therefore this doesn't undermine the comparison between these models here.

RMSE/MAE:

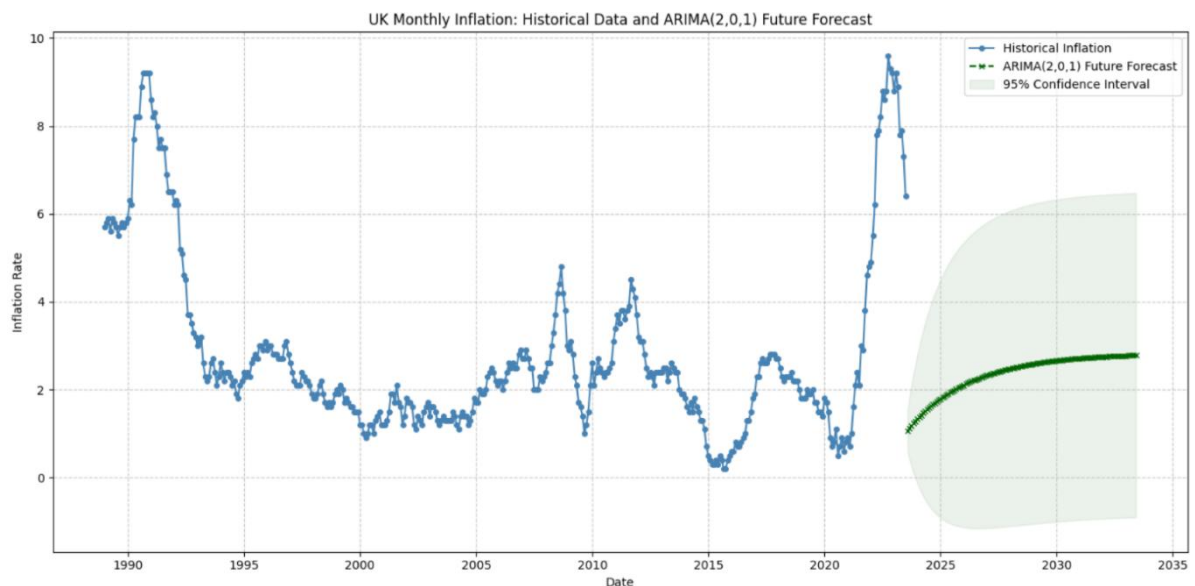
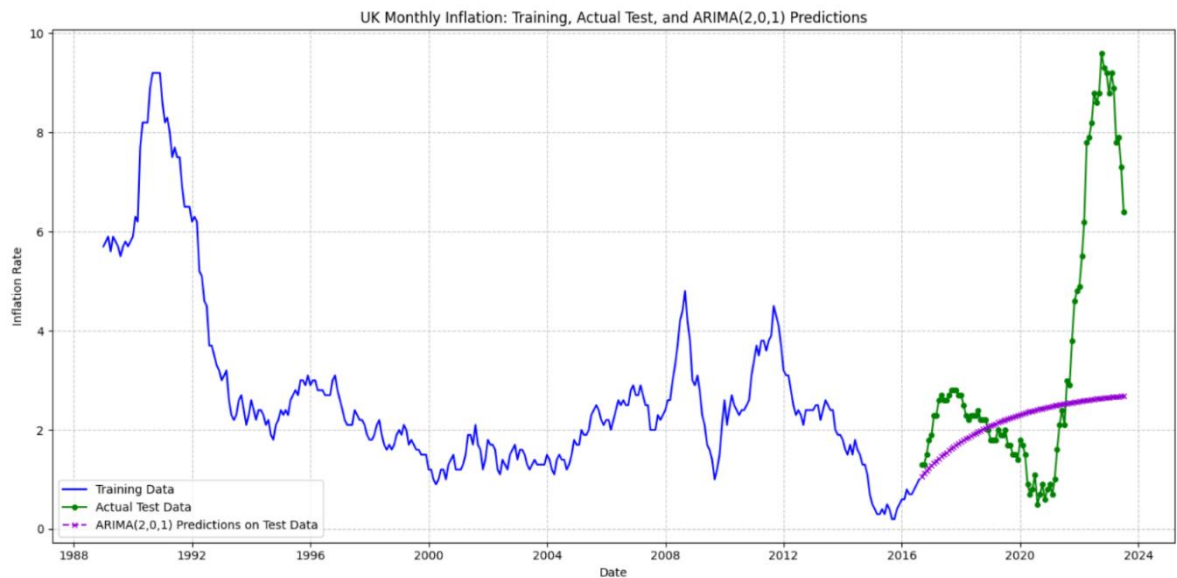
The RMSE and MAE are evaluation metrics that focus on prediction errors to evaluate how forecasts predicted by the model align with the actual observed values in the test set.

The MAE (mean absolute error) is the average absolute difference between forecasts and actual values (test set). This indicates how far on average the model's predictions are from the actual observed values irrespective of the direction of the error. The ARIMA (2,0,1) model shows an average forecasting error of 1.863 percentage points.

An advantage of the MAE is that it is easy to understand as well as easily interpretable with units in context of the original data. However, it treats all errors linearly. For example, an error of 5% is 2 times as bad as an error of 2.5%. When looking in the context of inflation, this may not be realistic as larger errors can be disproportionately costly.

The RMSE (root mean squared error) measure on the other hand is the square root of the squared differences of the predicted errors. Initially squaring errors gives more weight to larger errors than smaller ones. Then by taking the square root of this result, the data becomes interpretable in a way such as the MAE under the original data units. This may be preferred to the MAE as it can penalize larger errors more heavily than smaller ones. The RMSE for the ARIMA (2,0,1) is 2.764 percentage points. As seen in the table above, the RMSE values for all models exceed the MAE values which is expected as the RMSE is derived from squared errors.

Forecast graph:



Test set prediction:

From the visualization on the test set, the ARIMA (2,0,1) model evidently doesn't capture the true inflation values over this period. Instead, the smooth line models more of an average than the true fluctuations in the data.

However, the performance measures say otherwise as the model achieved an MAE of 1.863 percentage points and an RMSE of 2.764 percentage points which is small relative to the size of the fluctuations seen in the visual.

As the test period occurs during the global pandemic, this reflects the limitations of a univariate ARIMA model when faced with unprecedented macroeconomic shocks. At

this point, we can never know how accurate the forecasts would have been had the pandemic not happened.

Future forecast:

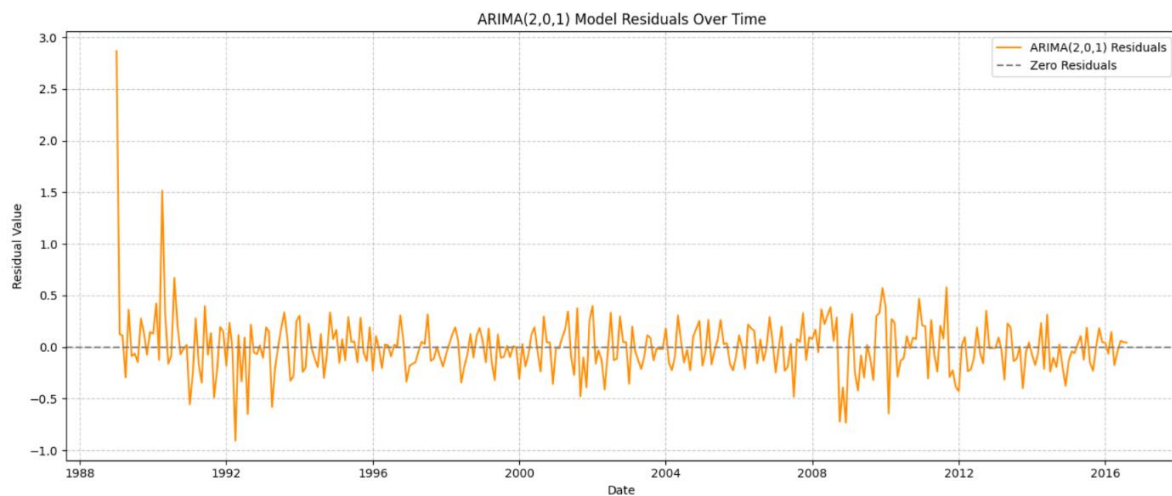
From the 95% confidence interval band, the model is cautious in predicting inflation due to large variances from unpredictable shocks which have persistence.

Hyndman and Athanasopoulos (2021) emphasize that prediction intervals for the ARIMA model depend on the assumption that residuals are uncorrelated and normally distributed. Under the failure of either assumption, prediction intervals may be incorrect. Additionally, previous historical patterns are assumed to continue into the forecast period which therefore can skew future predictions (Hyndman and Athanasopoulos, 2021).

Hyndman and Athanasopoulos (2021) further highlight that forecasting is typically unsuccessful with large outliers in the data, which can be replaced with estimates that remain consistent with the data pattern. However, replacing outliers without conducting reasoning towards their occurrence is unwise as these could provide useful information (Hyndman and Athanasopoulos, 2021). This is consistent with statements made from the BoE where outliers can actually suggest structural breaks and changing inflation dynamics, making room for new inflation forecasts (Baldwin et al., 2023).

The model predicts that that inflation will fall and rise slightly as it bottoms out over the forecast period with limited fluctuations at the long-run level. However, this should not be taken as concrete evidence that future inflation will follow this path and it is unrealistic that even minor fluctuations won't occur. As the model is built on historical patterns which don't account for structural changes, predictions are likely to be inaccurate over this volatile test period.

Residual plot:

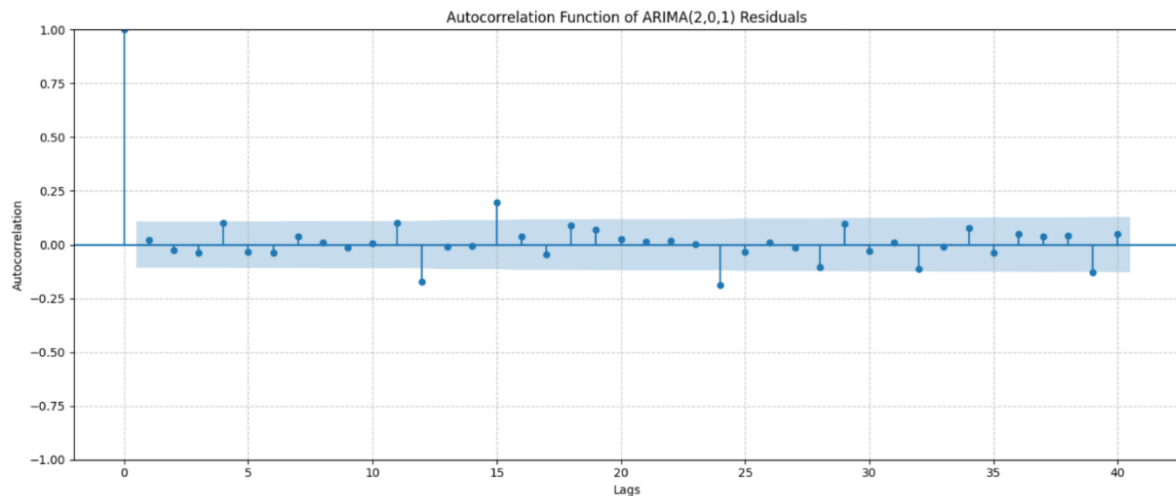


Residuals show the difference between the observed values and the predicted values from the model (the part of the data left unexplained by the model). If the ARIMA (2,0,1) model has successfully captured the AR, MA and stationarity of the time series, the residuals should behave like white noise; exhibiting no patterns.

Research suggests that the residuals should follow some conditions: a mean close to 0, constant variance with the spread being consistent over time, and no autocorrelation between residuals (as this indicates predictable structure left unexploited by the model).

Apart from a small anomaly in 1989, the residuals appear centred around 0 with a constant variance since values aren't shrinking or growing over time. There appears to be no autocorrelation between residuals as the residuals appear randomly scattered around 0. This provides evidence that the model has adequately captured the temporal structure in the series.

Residual ACF:



Furthermore, this can be verified with an ACF of the residuals. Under a good model fit, most correlation spikes should lie in the shaded confidence interval and there shouldn't be any significant autocorrelation, otherwise the model has likely unaccounted for a relationship in the data.

Whilst there is a spike at lag 0, this simply shows the correlation of the current value with itself. The absence of any other significant residual autocorrelation suggests that the ARIMA (2,0,1) model has captured the principal temporal dependence remaining in the series.

Limitations:

Past research has shown that the ARIMA model also comes with its disadvantages. Firstly, forecasting accuracy can deteriorate as the forecast horizon increases as past data is used for modelling the future, but structural relationships typically change over time. For example, Petrica et al. (2016) discusses that the ARIMA model struggles under outbreaks at irregular and unpredictable intervals with periods of volatility. Whilst constant variance is necessary for stationarity, periods of changing volatility may violate this assumption, potentially reducing the suitability of a simple ARIMA model (Petrică et al., 2016).

Furthermore, Hyndman and Athanasopoulos (2021) highlight problems with forecasting time series that are too long or too short. Firstly, they suggest the required sample size should increase with the number of parameters estimated and noise in the data. However, too many data points can lead to the difference between the model and the true process becoming obvious. They suggest fitting a model to the most recent observations as relationships don't have time to change substantially (Hyndman and Athanasopoulos, 2021).

Additionally, Hyndman and Athanasopoulos (2021) outline further problems with forecasts on test data where models are fitted with training data and then performance is evaluated on the test set. This is because, comparisons on test set data use different forecast horizons, averaging absolute or squared errors in measures such as MAE and RMSE, therefore combining results with different variances (Hyndman and Athanasopoulos, 2021).

The BoE highlights that the greatest weakness of the current forecasting framework is the difficulty when explaining persistence in UK inflation following unprecedented shocks. Although they consider forecast errors to be beneficial to reveal weaknesses in existing forecasts for prompting new policy judgements, high volatility breaks down forecasting frameworks. Therefore, central banks must rely on the judgement of their committee members to support policy decisions (Baldwin et al., 2023).

Discussion

The main findings of this report are that the ARIMA (2,0,1) model outperforms the other candidate ARIMA models for forecasting inflation from on the UK data over the 30-year period. The diagnostic tests suggest that the (2,0,1) model yields the least forecast errors. However, this isn't definitive evidence that the model will accurately forecast future inflation but instead it is an indicator that it provides the best performance against the competition.

Whilst literature confirms that ARIMA models have previously fit inflation forecasts successfully over test periods, this doesn't prove the model is 100% accurate for all time series data. Previous studies have identified that ARIMA models struggle with forecasting in the midst of economic shocks, showcased in ARIMA model predictions after Covid. This has been due to changing structural and statistical properties in data over time. This links to the problem of how far back the model should be fed data. Feeding the model too much historical data can fail to account for these structural changes however feeding the model data only from only a more recent time period is risky as shocks can skew even more recent underlying patterns. For example, using data from only 2008-2023 which has experienced the financial crisis, covid, and Ukraine war may skew the model predictions to favour higher inflation. Overall, as the variance of the permanent component from the inflation time series has changed over time, inflation continuously becomes harder to forecast posing a threat to policymakers.

If policy decisions are misinformed, this can lead to negative macroeconomic consequences as austerity measures implemented by central banks could make the economy worse off than if they were not to act at all. This suggests that there is increasingly more room for human judgement in policy decisions when quantitative methods fail. This raises the question of how much weight policymakers should place on quantitative forecasting models when their predictive performance may deteriorate during periods of significant structural change.

Although univariate forecasting models such as the ARIMA model incurs multiple limitations the econometric model is a more successful forecasting tool against the

alternative Phillips Curve method. Literature suggests there is room for machine learning methods in time series forecasting. Whilst this is still an area of growing research, this could be the future for increased forecasting accuracy that could be utilised by central banks alongside other tools. However, the trade-off between interpretability and accuracy means central banks must be informed on how much weight to put on this method to guide policy.

Conclusion

Overall, the ARIMA (2,0,1) was the best performing model against the alternative candidates for forecasting UK inflation. The ARIMA (2,0,1) model had the lowest AIC and BIC across all models at values of 13.257 and 32.283 percentage points respectively. Additionally, the average errors made in the prediction model captured by the RMSE and MAE were at 2.764 and 1.863 percentage points respectively which again outperformed other candidates as the predictions were closer to the actual outcomes for percentage change in inflation within the testing period. Furthermore, residual diagnostics via a residual plot and residual ACF plot provides evidence that the model has adequately captured the temporal structure in the series.

However, the model performance during the post pandemic period demonstrates the limitations of relying on historical dynamics for time series forecasts when structural shocks occur. Previous research has shown that using historical data from too far back in time can be problematic because there are likely structural breaks in the data over time from changes in the economic environment which can change the time series dynamics that were previously used for forecasting. For example, this is visible during the post Covid period that was also followed by the invasion of Ukraine in 2022. Trends have shown that demand and supply shocks have become more frequent over the last decade, creating volatility that has affected the time series and therefore skewed predictions. Therefore, this is why the ARIMA (2,0,1) future forecast doesn't appear to convey likely future fluctuations as it predicts a stable average and steers away from extreme values. However, it equally can't be confirmed that the suggested ARIMA model is a failed model for the future despite the fact it was unlucky to receive data over a period of high volatility.

To conclude, the primary takeaway is that the ARIMA model can be a useful component for the basis of an inflation forecasting framework but shouldn't be relied upon in isolation for monetary policy decisions due to the weakness of the model with its overreliance on historical data for future forecasts. The central bank should use the ARIMA model in combination with human insight and judgement amongst other tools to maintain a stable economy and macroeconomic conditions.

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